

Project Description

Managing budgets across different life needs—like home decor, event planning, or jewelry shopping—can be overwhelming due to the wide variety of products, platforms, and price ranges. PocketSmart AI addresses this challenge through a GenAI-powered, cross-platform recommendation system that delivers personalized, budget-based suggestions for products and services. Using Gemini 1.5 Flash Pro, PocketSmart AI analyzes user preferences, budgets, and contextual needs to generate curated recommendations from popular platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, and more. The platform helps users plan interior designs, parties, and even select matching jewelry for special occasions—all within their defined budgets. PocketSmart AI transforms budgeting into an intelligent, user-friendly experience through its smart planners, adaptive forms, and recommendation engine that bridges multiple categories and e-commerce ecosystems.

Scenario 1: Home Interior Planning with Smart Budget Allocation

Users begin by selecting the Home Budget Planner. After entering their budget, they specify room types (e.g., Living Room, Kitchen, Bedroom) and quantities for lights, ceiling fans, dining tables, etc. Gemini 1.5 Flash Pro processes this data and recommends cost-effective options for each category across platforms like IKEA and Amazon. Recommendations are balanced across functionality, style, and price, ensuring users can decorate their spaces beautifully without overspending.

Scenario 2: AI-Based Party Budget Planning

In the Party Planner, users input their total budget, guest count, event type, and venue details. Based on these inputs, the AI allocates the budget proportionally across catering, decoration, and entertainment, sourcing options from vendors listed on Swiggy, Zomato, and even accommodation services like OYO. The system tailors suggestions based on event type (e.g., birthday, corporate, wedding), helping users organize a successful event without financial guesswork.

Scenario 3: Jewelry Recommendations for Occasions

The Jewelry Budget Planner allows users to enter a budget, select the occasion, and define style preferences. Users can optionally upload an outfit image for the AI to consider color coordination and aesthetics. Gemini 1.5 Flash Pro analyzes the data and offers jewelry options from platforms like Amazon and Flipkart, tailored to occasion and style. This ensures that users receive elegant, matching jewelry suggestions that fit both their personal style and financial plan.

Architecture Overview

PocketSmart AI is an intelligent, modular platform designed to generate personalized, budget-aware recommendations across domains like home interiors, party planning, and jewelry selection. It integrates a lightweight Flask backend with Google’s Gemini 1.5 Flash Pro, a powerful multimodal AI model that processes user inputs (text, numbers, and images) to deliver smart recommendations.

Core Technologies

- **Flask:** Backend service layer for request handling, routing, and session control.
- **Gemini 1.5 Flash Pro:** AI model used for understanding budget context, analyzing images, generating structured suggestions, and recommending platform-specific products.
- **Third-party APIs (Amazon, Flipkart, IKEA, Zomato, etc.):** Product and service data sourcing.
- **Frontend (HTML/CSS + JS):** Interactive user interface for input collection and AI output display.

Component-Wise Architecture

Component	Description
Frontend UI	Web interface built with HTML/CSS allows users to input budgets, preferences, and images. Displays tailored AI-driven recommendations for each category.
Flask Backend	Manages user sessions, routes (/generate-home, /generate-party, /generate-jewelry), and handles communication between UI and Gemini AI.
Gemini AI Layer	Processes inputs using Gemini 1.5 Flash Pro to:

Pre-requisites

